higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CONCRETE STRUCTURES
NQF LEVEL 3

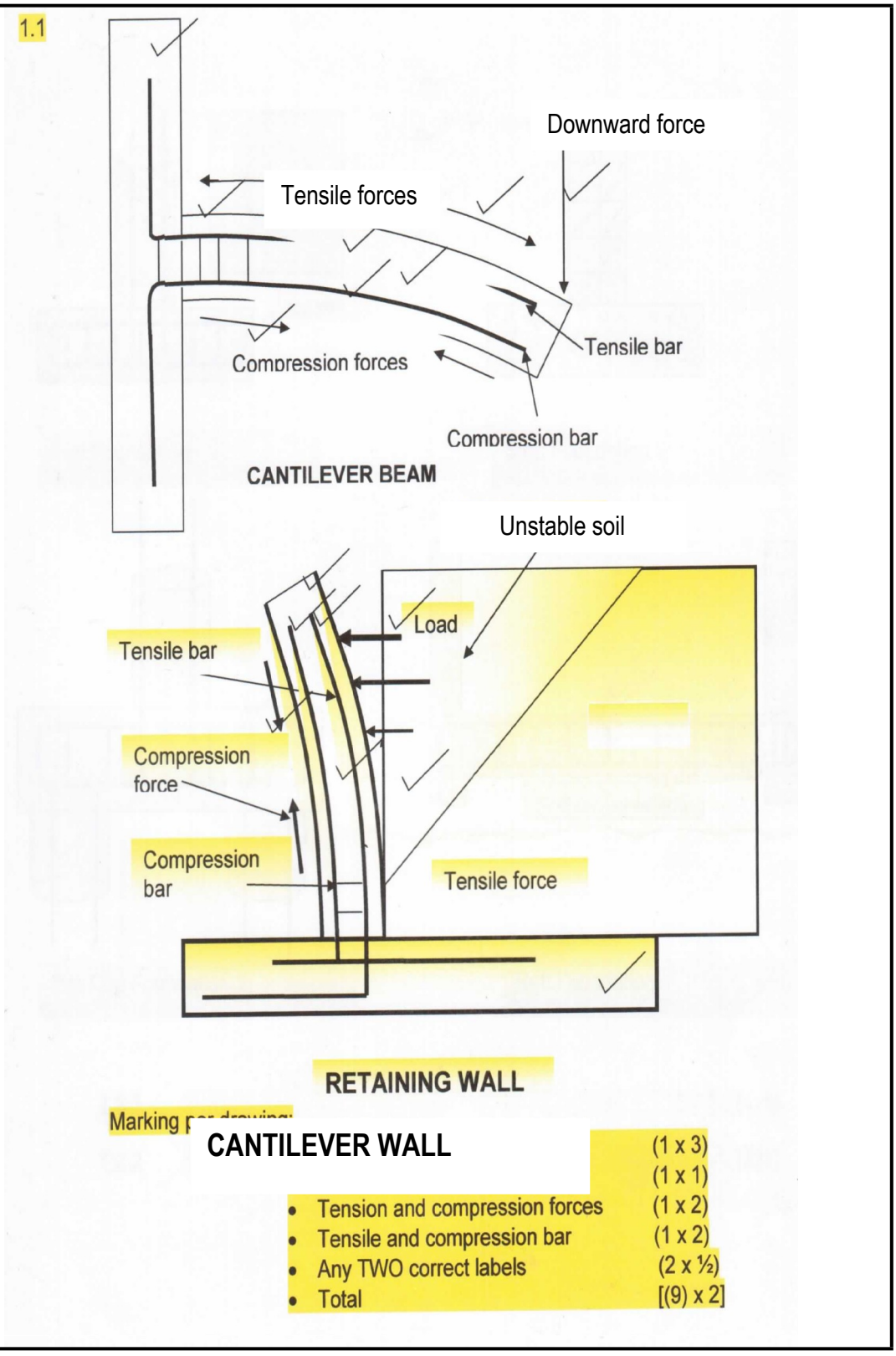
11 MARCH 2019

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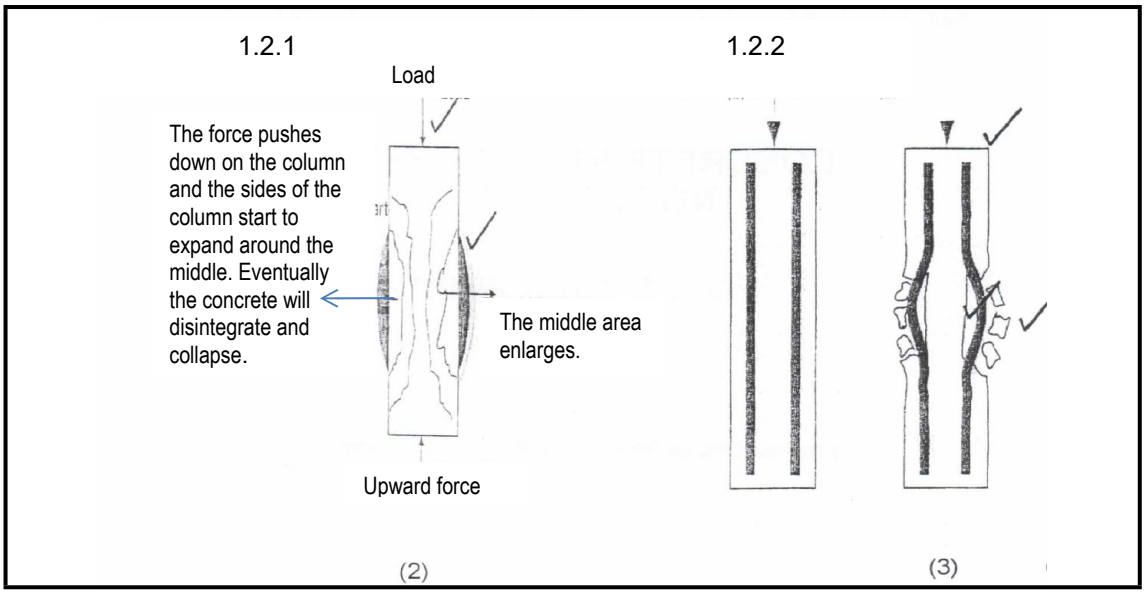
QUESTION 1: THE THEORY AND USE OF REINFORCED CONCRETE

1.1



(18)

1.2



(5)

1.3
- Bond failure
- Bearing failure
- Corrosion
- Restraint
- Settlement
- Thermal shrinkage

(Any 2 × 1) (2)

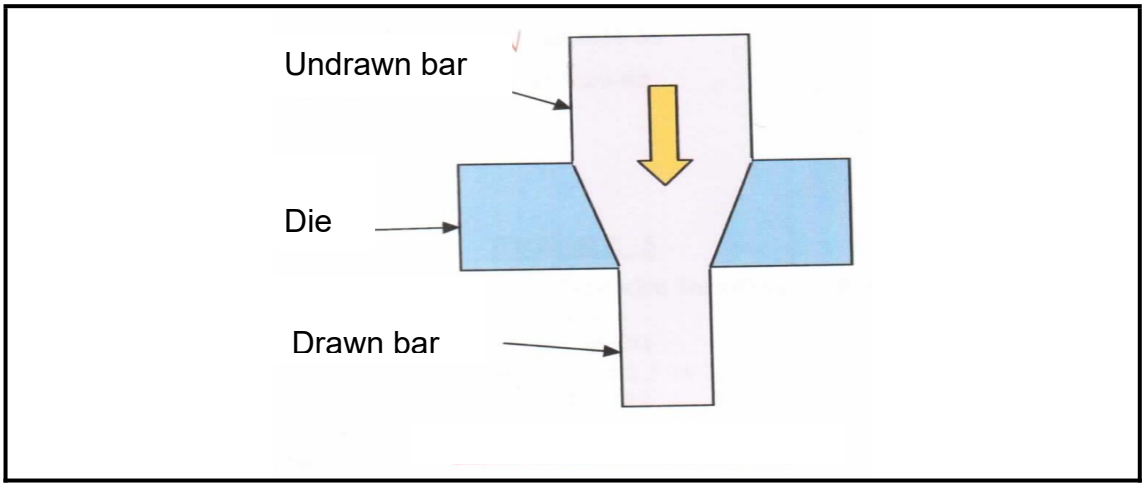
[25]

QUESTION 2: REINFORCED CONCRETE IN CONSTRUCTION

2.1
2.1.1 B
2.1.2 C
2.1.3 D
2.1.4 E
2.1.5 A

(5 × 1) (5)

2.2



BAR-DRAWING PROCESS (4)

2.3 R✓ – Refers to plain, round bars with a strength of 250 MPa✓
Y✓ – Refers to high-yield deformation steel bars with a strength of 450 MPa✓
Z✓ – Refers to types of steel not covered by R or Y✓
SM✓ – Refers to standard mesh (high-tensile wire with a strength of 485 MPa)✓
DM✓ – Refers to design mesh (high-tensile wire with a strength of 485 MPa)✓
(Any 4 × 2) (8)

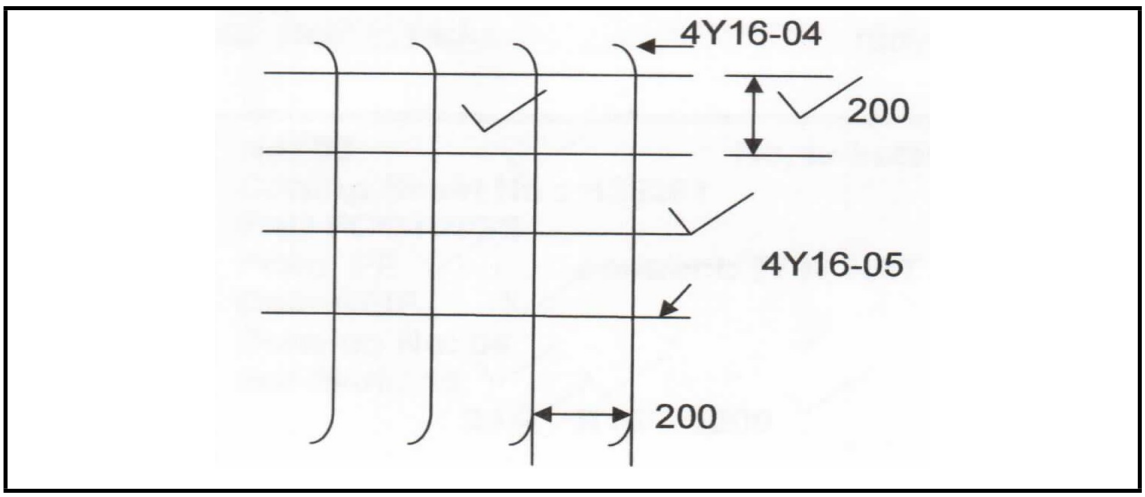
2.4 8 mm, 10 mm, 12 mm, 16 mm, 20 mm, 25 mm, 32 mm (Any 3 × 1) (3)

2.5
- Fibres should be much stiffer than matrix.
- Fibre content by volume must be adequate.
- There must be a good fibre matrix bond.
- Fibre must have a high aspect ratio.
- Fibre length must be sufficient. (5 × 1) (5)

[25]

QUESTION 3: REINFORCING DOCUMENTATION

3.1



(4)

3.2 3.2.1 4 – number of bars,✓ 20 – diameter✓ and Y – high-yield steel✓
(Any 3 × 1) (3)

3.2.2 5 – number of bars,✓ 8 – diameter✓ and R - mild steel✓(Any 3 × 1) (3)

3.3 3.3.1 Continuous thick line
3.3.2 Continuous thin line
3.3.3 Broken line
3.3.4 Chain dotted
3.3.5 Shaded portion or hatched areas
(5 × 1) (5)

3.4 (a) T1 = 1st layer from top
(b) T2 = 2nd layer from top
(c) B1 = 1st layer from bottom
(d) B2 = 2nd layer from bottom

(4 × 1) (4)

3.5 3.5.1 Straight lengths (including reinforcement for subsequent bending)
3.5.2 ±5 mm
3.5.3 Less than 1 mm
3.5.4 ±5 mm
3.5.5 More than 2 m
3.5.6 ±5 mm

(6 × 1) (6)
[25]

QUESTION 4: FABRICATION OF REINFORCING STEEL

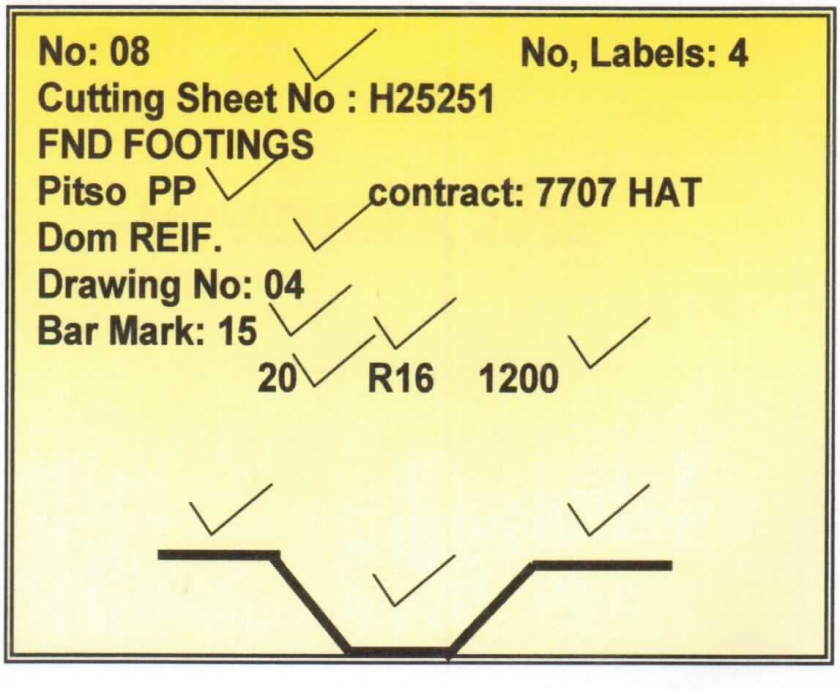
4.1 (a) Receiving
(b) Cutting
(c) Bending
(d) Administration
(e) Dispatch (5 × 1) (5)

4.2

TYPES OF DOCUMENTS	WHAT IS THEIR PURPOSE?	WHO USE THEM?	WHAT INFORMATION IS GIVEN?
Cutting and bending sheet	Tell the steel cutter and steel bender how the steel for a particular job must be cut and bent (2)	Yard foreman, steel cutter, steel bender people who load the bent steel onto the delivery trucks, driver of the delivery truck (1)	Order and job number, diameter, length, shapes and quality of steel, for whom and what the steel is for (1)
Bar label	Label bars of steel. Each label gives specific information about a bar of steel (1)	Steel cutter and steel bender, people who load the bent steel onto the truck, people using the steel on site (1)	Cutting and bending number, customer details, drawing reference, bar mark, details, shape, information, including diameters, code, lengths (1)
Delivery note	Give information about the steel that will be delivered to a customer/site (1)	Driver who delivers the steel, person who receives the steel delivery (1)	Total mass of the types of steel delivered, shape codes of the different steel bars, customer's name, delivery address, order number, date of delivery (1)

(10)

4.3



(10)
[25]

TOTAL: 100

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